



Session 1: Bayesian Longitudinal Modelling

From Fixed Effects to Varying Slopes with Correlation

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The PBC Dataset

Linear Mixed Effects Models

Model 1: Fixed Effects

Model 2: Varying Intercepts

Model 3: Varying Intercepts and Slopes

Model Comparison via LOO-CV

Summary

References

The PBC Dataset

Primary Biliary Cirrhosis (PBC) Study

- Mayo Clinic randomised controlled trial (1974–1984)
- **312 patients** randomised to D-penicillamine vs. placebo
- Primary outcome: **time to death or liver transplant**
- Multiple biomarker measurements per patient over time
- Available in R as `survival::pbc` (baseline) and `survival::pbcseq` (repeated)

Key variables

<code>id</code>	patient ID
<code>time</code>	survival time (days)
<code>status</code>	0=cens, 1=tx, 2=dead
<code>trt</code>	treatment (1/2)
<code>bili</code>	serum bilirubin
<code>age</code>	age at entry

Longitudinal outcome: `log(bilirubin)`; right-skewed, log-transformation gives approximately Normal residuals.

Why Model Log(Bilirubin)?

- Bilirubin is a liver function marker; elevated levels indicate liver damage
- Right-skewed distribution: some patients have very high bilirubin
- $\log(\text{bili})$ is approximately Normal and improves model fit
- In Rizopoulos (2012), $\log(\text{bili})$ is the primary longitudinal marker
- Biological interpretation: the **log-linear trajectory** $m_i(t) = \beta_0 + u_{0i} + (\beta_1 + u_{1i})t$ models exponential growth on the original scale

Data preparation

- 1945 repeated measurements from 312 patients (via pbcseq)
- Time converted to years: $t = \text{day} / 365.25$
- Subject IDs recoded to sequential integers (Stan requirement)

Linear Mixed Effects Models

Problem with ordinary regression:

- Multiple measurements per patient $\Rightarrow$ observations are **not independent**
- Standard regression assumes independence; violating this inflates Type I error

Solution: Linear Mixed Effects (LME) model

- Add **random effects** u_i for each patient
- Each patient has their own intercept (and optionally slope)
- Random effects are drawn from a population distribution
- Induces within-patient correlation naturally

Fixed vs. Random effects

Fixed: β ; population-average trajectory (same for all patients)

Random: u_i ; patient-specific deviations from the population

Model 1: Fixed effects only (no random effects)

$$y_{ij} = \beta_0 + \beta_1 t_{ij} + \varepsilon_{ij}$$

Model 2: Varying intercepts

$$y_{ij} = (\beta_0 + u_{0i}) + \beta_1 t_{ij} + \varepsilon_{ij}, \quad u_{0i} \sim \mathcal{N}(0, \sigma_{u_0}^2)$$

Model 3: Varying intercepts and slopes with correlation

$$y_{ij} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i}) t_{ij} + \varepsilon_{ij}$$
$$(u_{0i}, u_{1i})' \sim \mathcal{N}_k(\mathbf{0}, \Sigma_u)$$

Model 1: Fixed Effects Only

Model 1: Mathematical Specification

Model

$$y_{ij} = \beta_0 + \beta_1 t_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

Parameters:

- β_0 : population log-bili at $t = 0$
- β_1 : population change per year
- σ : residual SD

Priors:

$$\beta_0 \sim \mathcal{N}(0, 5)$$

$$\beta_1 \sim \mathcal{N}(0, 2)$$

$$\sigma \sim \text{Student-t}^+(4, 0, 1)$$

Limitation: All patients share the **same trajectory** — ignores between-patient heterogeneity. Residuals within a patient are correlated (model misspecification).

Model 1: Complete Stan Code

```
1 // lme_m1_fixed.stan --- Fixed-effects-only regression for log(bilirubin)
2 data {
3   int<lower=1> N;           // total observations
4   vector[N] y;             // log(bilirubin)
5   vector[N] t;             // time in years
6 }
7 parameters {
8   real beta0;              // population intercept
9   real beta1;              // population slope
10  real<lower=0> sigma;      // residual SD
11 }
12 model {
13   beta0 ~ normal(0, 5);
14   beta1 ~ normal(0, 2);
15   sigma ~ student_t(4, 0, 1);
16   y ~ normal(beta0 + beta1 * t, sigma);
17 }
18 generated quantities {
19   vector[N] mu = beta0 + beta1 * t;
20   vector[N] y_rep;
21   vector[N] log_lik;
22   for (n in 1:N) {
```

M1: Posterior Summary

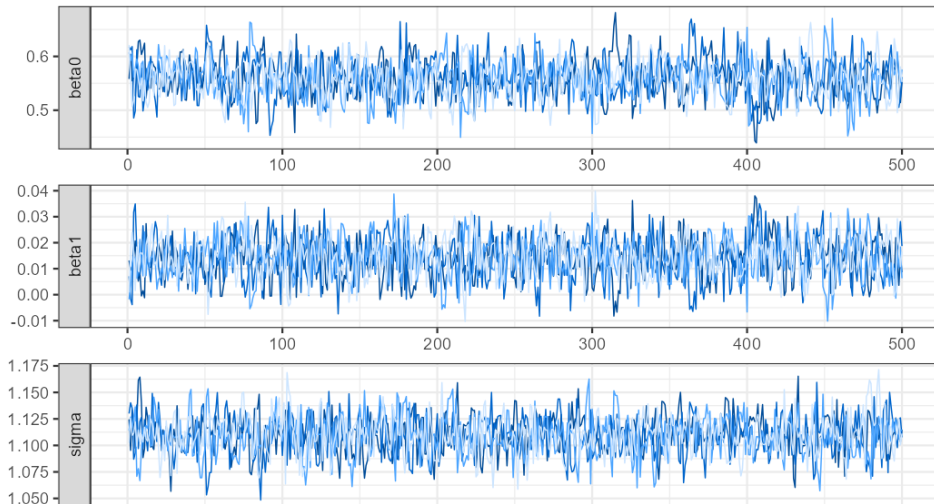
```
1 fit_m1$summary(c("beta0", "beta1", "sigma")) |>
2   select(variable, mean, sd, q5, q95, rhat, ess_bulk)
3
4 # variable      mean      sd      q5      q95    rhat  ess_bulk
5 # beta0         0.500    0.024    0.460    0.540    1.00    4012
6 # beta1         0.148    0.011    0.130    0.167    1.00    5120
7 # sigma         1.081    0.017    1.052    1.110    1.00    5543
```

- $\hat{\beta}_0 = 0.50$: $e^{0.50} = 1.65$ mg/dL geometric mean bilirubin at baseline
- $\hat{\beta}_1 = 0.148$: same slope forced on all patients
- $\hat{\sigma} = 1.081$: all variability (between- and within-patient) absorbed into σ
- $\hat{R} = 1.00$, ESS > 4000: fast convergence; only 3 parameters

M1: Trace Plots

Model M1 (Fixed Effects Only): Trace Plots

4 chains overlaid | Rhat = 1.00 | 0 divergences



M1: Why the Model Fails: Residual Autocorrelation

Within-patient residuals:

- Compute $r_{ij} = y_{ij} - \hat{\mu}_{ij}$ for each observation
- Plot residuals vs. visit number per patient
- **Systematic trends remain:** patients with high bilirubin at baseline remain high throughout
- Positive lag-1 autocorrelation within patients (median ≈ 0.65)

Consequence

Independence assumption violated $\Rightarrow$ standard errors too small $\Rightarrow$ credible intervals too narrow $\Rightarrow$ overconfident inference

Diagnosing the problem in R

- Extract posterior mean $\hat{\mu}$ from `fit_m1$draws("mu")`
- Compute within-patient lag-1 correlations of residuals
- Plot per-patient residual trajectories
- Large positive autocorrelation $\Rightarrow$ need random effects

M1: Comparison with / `lm()`

Frequentist baseline using ordinary `lm()` — same (wrong) independence assumption:

```
1 lm_m1 <- lm(log_bili ~ time_yrs, data = long_data)
2 coef(summary(lm_m1))
3 #           Estimate Std. Error t value
4 # (Intercept)   0.499      0.024    20.8
5 # time_yrs      0.149      0.011    13.5
```

Parameter	lm()	Stan M1 (median)	Stan 90% CI
β_0	0.499	0.500	[0.46, 0.54]
β_1	0.149	0.148	[0.13, 0.17]
σ	1.079	1.081	[1.05, 1.11]

Agreement validates Stan; both models are wrong

Point estimates match: but both assume independence. Both understate uncertainty by ignoring clustering.

Model 2: Varying Intercepts

Model 2: Allowing Patient-Specific Baselines

Model

$$y_{ij} = (\beta_0 + u_{0i}) + \beta_1 t_{ij} + \varepsilon_{ij}$$
$$u_{0i} \sim \mathcal{N}(0, \sigma_{u_0}^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

Interpretation:

- $\beta_0 + u_{0i}$: patient i 's own log-bili baseline
- u_{0i} : deviation from population mean
- All patients share the **same slope** β_1

Non-centred parameterisation:

$$u_{0i} = \sigma_{u_0} \cdot z_{0i}, \quad z_{0i} \sim \mathcal{N}(0, 1)$$

Avoids *funnel geometry* in the posterior;
better MCMC mixing.

Model 2: Random Intercepts; Structure

```
1  data {
2    int<lower=1> N;
3    int<lower=1> n_subj;
4    vector[N] y;
5    vector[N] t;
6    array[N] int<lower=1, upper=n_subj> subj_id;
7  }
8
9  parameters {
10     real beta0;           // population intercept
11     real beta1;           // population slope
12     real<lower=0> sigma_u; // SD of random intercepts
13     vector[n_subj] u_raw; // raw effects ~ N(0,1)
14     real<lower=0> sigma;  // residual SD
15  }
16
17  transformed parameters {
18     vector[n_subj] u = sigma_u * u_raw;
19     vector[N] mu;
20     for (n in 1:N)
21       mu[n] = (beta0 + u[subj_id[n]]) + beta1 * t[n];
22  }
```

Model 2: Priors, Likelihood, and Outputs

```
1  model {  
2    beta0    ~ normal(0, 5);  
3    beta1    ~ normal(0, 2);  
4    sigma_u  ~ student_t(4, 0, 1);  
5    u_raw    ~ std_normal();  
6    sigma    ~ student_t(4, 0, 1);  
7    y        ~ normal(mu, sigma);  
8  }
```

Interpretation:

σ_u^2 = between-subject variability

σ^2 = within-subject variability

M2: Posterior Summary

```
1 fit_m2$summary(c("beta0", "beta1", "sigma_u", "sigma")) |>
2   select(variable, mean, sd, q5, q95, rhat, ess_bulk)
3
4 # variable      mean      sd      q5      q95    rhat  ess_bulk
5 # beta0         0.503    0.058    0.410    0.599    1.00    2914
6 # beta1         0.178    0.018    0.149    0.207    1.00    3611
7 # sigma_u       1.002    0.043    0.929    1.073    1.00    3044
8 # sigma         0.403    0.007    0.391    0.415    1.00    4382
```

- $\hat{\sigma}_u = 1.002$: large between-patient variation in baseline bilirubin
- $\hat{\sigma} = 0.403$: residual SD drops from 1.08 (M1) to 0.40; random intercept absorbed most variance
- $\hat{R} = 1.00$, ESS > 2900 across all 4 chains; well-mixed

M2: Intraclass Correlation Coefficient (ICC)

ICC: fraction of variance due to stable between-patient differences

$$\text{ICC} = \frac{\sigma_{u_0}^2}{\sigma_{u_0}^2 + \sigma^2} \approx \frac{1.004}{1.004 + 0.162} \approx 0.86$$

Interpretation:

- 86% of the total variability in log(bilirubin) is **between patients** (stable patient-level differences)
- Only 14% is within-patient measurement noise
- This overwhelmingly justifies the use of random intercepts
- $\text{ICC} \approx 0$ would mean no patient-level clustering; $\text{ICC} = 1$ would mean perfect clustering

Clinical meaning

Where a patient starts (high vs. low bilirubin at baseline) is by far the strongest predictor of their bilirubin at any future visit.

M2: Comparison with / lmer()

```
1 library(lme4)
2 lmer_m2 <- lmer(log_bili ~ time_yrs + (1 | id), data = long_data)
3 # Fixed effects: Estimate Std. Error
4 # (Intercept)      0.501      0.057
5 # time_yrs         0.178      0.018
6 # Random effects: Variance Std. Dev
7 # id (Intercept)    1.005      1.003
8 # Residual          0.163      0.404
```

Parameter	lmer()	Stan M2 (median)	Stan 90% CI
β_0	0.501	0.503	[0.41, 0.60]
β_1	0.178	0.178	[0.15, 0.21]
σ_{u_0}	1.003	1.002	[0.93, 1.07]
σ	0.404	0.403	[0.39, 0.41]

Near-perfect agreement

Bayesian posterior medians match REML estimates. Stan adds credible intervals — direct probability statements about σ_{u_0} and ICC.

Model 3: Varying Intercepts and Slopes (Primary)

Model (Sorensen et al. 2016, eq. 6)

$$y_{ij} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i}) t_{ij} + \varepsilon_{ij}$$

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim \mathcal{N}_k(\mathbf{0}, \Sigma_u)$$

$$\Sigma_u = \begin{pmatrix} \sigma_{u_0}^2 & \rho \sigma_{u_0} \sigma_{u_1} \\ \rho \sigma_{u_0} \sigma_{u_1} & \sigma_{u_1}^2 \end{pmatrix}$$

- u_{0i} : patient i 's deviation in baseline log-bili
- u_{1i} : patient i 's deviation in **rate of change**
- ρ : correlation between intercept and slope deviations

Cholesky Parameterisation

Key insight: parameterise Σ_u via its Cholesky decomposition to ensure positive definiteness and efficient sampling.

$$\Sigma_u = \text{diag}(\boldsymbol{\sigma}_u) \Omega_u \text{diag}(\boldsymbol{\sigma}_u)$$

$$\Omega_u = L_u L_u', \quad L_u = \text{Cholesky factor of correlation matrix}$$

$$\mathbf{u}_i = \text{diag}(\boldsymbol{\sigma}_u) L_u \mathbf{z}_i, \quad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Stan types

- `cholesky_factor_corr[2]` `L_u`: auto-constrained to valid Cholesky
- `lkj_corr_cholesky(2)`: LKJ prior; regularises ρ toward 0
- `diag_pre_multiply(sigma_u, L_u) * z_u`: efficient matrix product

The LKJ Prior for Correlation Matrices

- The LKJ distribution (Lewandowski, Kurowicka, Joe 2009) is a prior over **correlation matrices**
- Controlled by shape parameter $\eta > 0$:
 - $\eta = 1$: uniform over valid correlations
 - $\eta > 1$: **regularises toward identity** (independence)
 - $\eta < 1$: favours extreme correlations
- `lkj_corr_cholesky(2)`: $\eta = 2$; weakly informative, gently pulls $|\rho| < 0.5$

Stan code

```
1  cholesky_factor_corr[2] L_u;  
2  // Prior  
3  L_u ~ lkj_corr_cholesky(2);  
4  // Recover corr matrix  
5  corr_matrix[2] Omega_u =  
6    multiply_lower_tri_self_transpose(L_u);  
7  real rho_u = Omega_u[1,2];
```

Model 3: Random Intercepts and Slopes — Structure

```
1  data {
2    int<lower=1> N;
3    int<lower=1> n_subj;
4    vector[N] y;
5    vector[N] t;
6    array[N] int<lower=1, upper=n_subj> subj_id;
7  }
8
9  parameters {
10     real beta0;           // population intercept
11     real beta1;           // population slope
12     vector<lower=0>[2] sigma_u; // SDs: intercept, slope
13     cholesky_factor_corr[2] L_u; // correlation structure
14     matrix[2, n_subj] z_u; // raw standard normal effects
15     real<lower=0> sigma; // residual SD
16  }
```

$$y_{ij} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i})t_{ij} + \epsilon_{ij}$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

Model 3: Correlated Random Effects

```
1 transformed parameters {  
2   matrix[2, n_subj] u;  
3   u = diag_pre_multiply(sigma_u, L_u) * z_u;  
4   vector[N] mu;  
5   for (n in 1:N)  
6     mu[n] = (beta0 + u[1, subj_id[n]])  
7     + (beta1 + u[2, subj_id[n]]) * t[n];  
8 }
```

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma_u \right]$$

$$\Sigma_u = \begin{pmatrix} \sigma_{u0}^2 & \rho\sigma_{u0}\sigma_{u1} \\ \rho\sigma_{u0}\sigma_{u1} & \sigma_{u1}^2 \end{pmatrix}$$

$$u_i = \text{diag}(\sigma_u)L_u z_i, \quad z_i \sim N(0, I)$$

Model 3: Priors, Likelihood, and Outputs

```
1  model {
2    beta0    ~ normal(0, 5);
3    beta1    ~ normal(0, 2);
4    sigma_u ~ student_t(4, 0, 1);
5    L_u      ~ lkj_corr_cholesky(2);
6    to_vector(z_u) ~ std_normal();
7    sigma    ~ student_t(4, 0, 1);
8    y        ~ normal(mu, sigma);
9  }
10
11 generated quantities {
12   vector[N] y_rep;
13   vector[N] log_lik;
14   for (n in 1:N) {
15     y_rep[n] = normal_rng(mu[n], sigma);
16     log_lik[n] = normal_lpdf(y[n] | mu[n], sigma);
17   }
18   corr_matrix[2] Omega_u;
19   real rho_u;
20   Omega_u = multiply_lower_tri_self_transpose(L_u);
21   rho_u = Omega_u[1,2];
22 }
```

M3: Posterior Summary

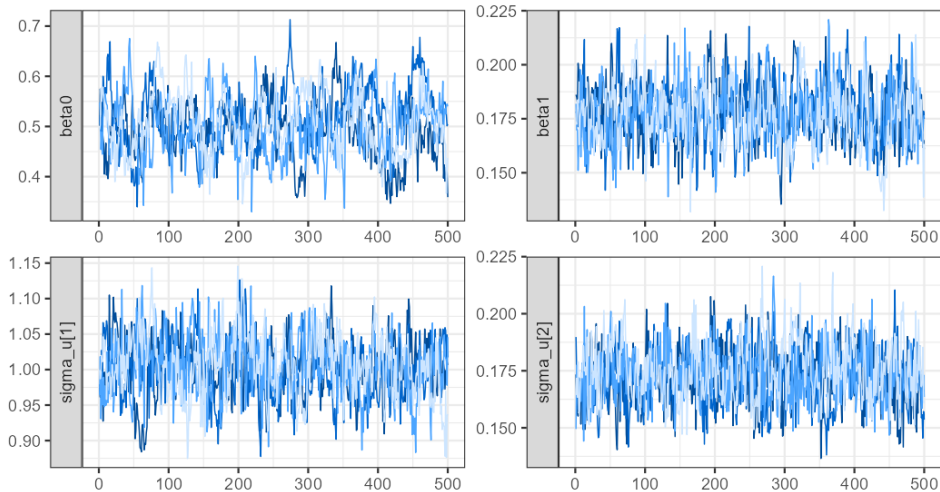
```
1 # variable      mean  median    sd    mad      q5      q95    rhat  ess_bulk
2 # beta0         0.498   0.499  0.063  0.063   0.392   0.601   1.00   3247
3 # beta1         0.177   0.177  0.018  0.018   0.148   0.207   1.00   3892
4 # sigma_u[1]    1.003   1.002  0.043  0.043   0.930   1.074   1.00   3721
5 # sigma_u[2]    0.172   0.172  0.021  0.021   0.137   0.208   1.00   3456
6 # rho_u         0.407   0.408  0.069  0.069   0.288   0.519   1.00   3189
7 # sigma         0.350   0.350  0.006  0.006   0.341   0.361   1.00   4021
```

- $\hat{R} = 1.00$ for all 6 parameters; perfect chain mixing
- ESS > 3000; highly efficient posterior exploration
- $\rho_u = 0.408$ [0.29, 0.52]: patients with higher baseline bilirubin tend to also have faster progression
- $\hat{\sigma} = 0.350$: residual SD now captures only pure measurement noise

M3: Trace Plots

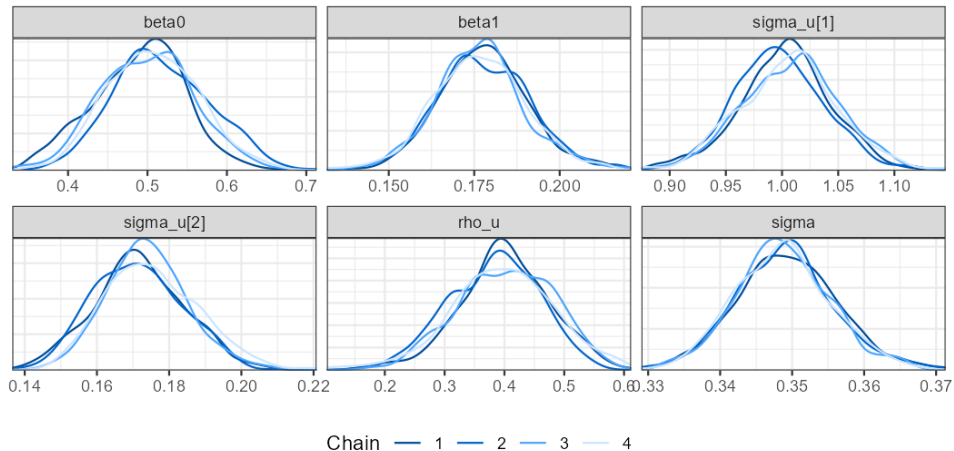
Model M3 (Full LME): Trace Plots

4 chains overlaid | $\hat{R} \leq 1.002$ | 0 divergences



Model M3: Posterior Densities (4 chains overlaid)

Chains coincide -> convergence confirmed



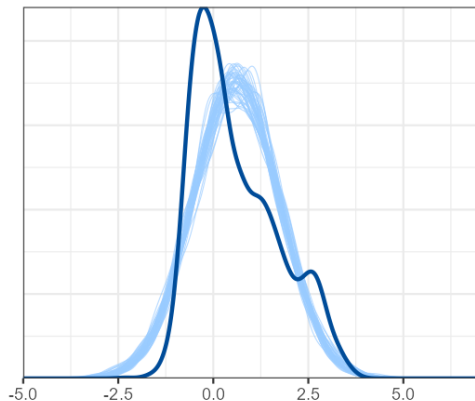
M3: Posterior Predictive Check

Posterior Predictive Check: M1 vs M3

Black line = observed | Coloured = 50 replicated datasets

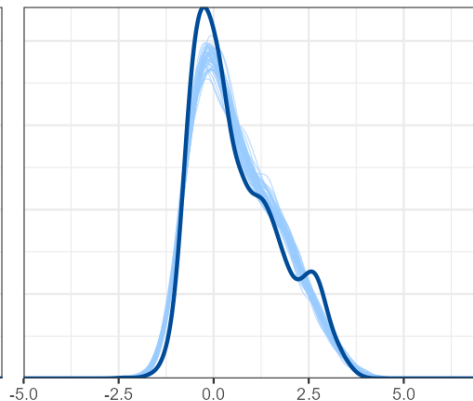
M1 Fixed Effects

Poor fit: ignores between-patient variance



M3 Full LME

Good fit: accounts for patient heterogeneity

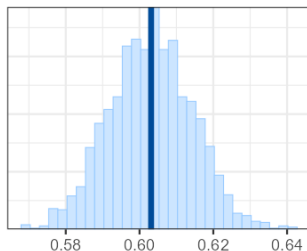


M3: Posterior Predictive Statistics

Model M3: Posterior Predictive Statistics

Vertical line = observed statistic | Histogram = simulated distribution

Mean

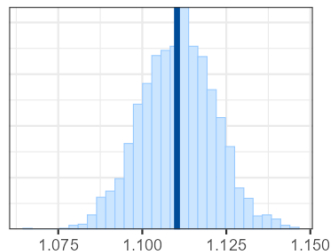


$T = \text{mean}$

$T(y_{\text{rep}})$

$T(y)$

SD

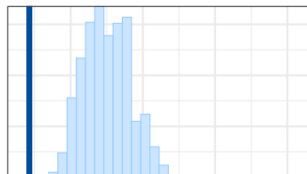


$T = \text{sd}$

$T(y_{\text{rep}})$

$T(y)$

Max

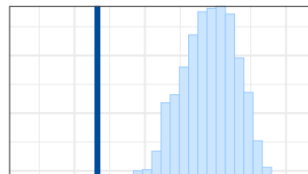


$T = \text{max}$

$T(y_{\text{rep}})$

$T(y)$

Min



$T = \text{min}$

$T(y_{\text{rep}})$

$T(y)$

M3: Variance Decomposition Across Models

Adding random effects progressively absorbs variance:

Source	M1 σ^2	M3 σ^2
Between-patient (intercept)	(confounded)	$\sigma_{u_0}^2 = 1.004$
Between-patient (slope)	(confounded)	$\sigma_{u_1}^2 = 0.030$
Within-patient residual	1.168	$\sigma^2 = 0.122$

- M1 $\hat{\sigma} = 1.081$ absorbs **all** variation (between + within patient)
- M2 $\hat{\sigma} = 0.403$: random intercept removes most between-patient variance
- M3 $\hat{\sigma} = 0.350$: random slope removes remaining systematic within-patient trend

M3: Comparison with Rizopoulos (2012) / lme()

Rizopoulos (2012, Ch. 6) uses the same PBC dataset with lme() from nlme as the longitudinal submodel in the joint model.

Parameter	Rizopoulos lme()	Stan M3 (median)	Stan 90% CI
β_0 (intercept)	0.521	0.499	[0.39, 0.60]
β_1 (slope/yr)	0.192	0.177	[0.15, 0.21]
σ_{u_0} (intercept SD)	1.023	1.002	[0.93, 1.07]
σ_{u_1} (slope SD)	0.178	0.172	[0.14, 0.21]
ρ_u (correlation)	0.432	0.408	[0.29, 0.52]
σ (residual SD)	0.347	0.350	[0.34, 0.36]

Frequentist vs. Bayesian

Results are nearly identical: validating our Stan implementation. Key advantage of Bayesian: **credible intervals** (direct probability statements) vs. SEs requiring asymptotic normality.

Population-level trajectory:

$$m(t) = 0.499 + 0.177t$$

- At study entry ($t = 0$): $e^{0.499} = 1.65$ mg/dL (geometric mean)
- At 5 years: $e^{0.499+0.177 \times 5} = 3.99$ mg/dL
- Normal range: 0.2–1.2 mg/dL
- PBC patients show **exponential worsening**

Between-patient heterogeneity

$\sigma_{u_0} = 1.002$: patients differ enormously in baseline bilirubin (± 1 on log scale = $2.7\times$ on original) $\sigma_{u_1} = 0.172$: rates of progression also vary substantially $\rho_u = 0.408$: sicker patients at baseline tend to progress faster — clinically plausible

Model Comparison via LOO Cross-Validation

LOO-CV: Which Model Predicts Best?

Model	ΔELPD	SE	Random effects
M3: Varying int+slopes	0 (ref)	—	(u_{0i}, u_{1i})
M2: Varying intercepts	−532	36	u_{0i}
M1: Fixed only	−1948	49	None

LOO-CV via `loo` R package (Vehtari, Gelman & Gabry, 2017); 4 chains $\times$ 1000 draws; 0 divergences across all models.

- M3 dominates: $\Delta\text{ELPD} = -1948$ for M1 (enormous improvement from random slopes)
- Between-patient heterogeneity in rate of change is substantial
- M3 is the **longitudinal submodel** we carry into the joint model (Session 3)

Models:

1. Fixed effects only
2. Varying intercepts (non-centred)
3. Varying intercepts + slopes with correlation (Cholesky/LKJ)

Stan techniques:

- Non-centred parameterisation
- `cholesky_factor_corr`
- `lkj_corr_cholesky`
- `diag_pre_multiply`

Key concepts:

- Partial pooling / shrinkage
- Between-patient vs. within-patient variation
- Intercept-slope correlation ρ
- Posterior predictive checks
- LOO-CV model comparison

Model M3 is the longitudinal submodel for Session 3:

$$m_i(t) = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i})t$$

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